

Residential Electric Vehicle Charging Station using the Internet of Things

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Abstract—The available options for Electric Vehicles (EVs) charging need costly installation, restricted accessibility, and no remote management features. These obstacles could deter homeowners from switching to EVs and slow the pace at which EVs are adopted in the home sector. To solve these concerns, this work aims to create a cost-effective, user-friendly EV charging station (EVCS) that facilitates control, remote monitoring, and payment handling. Using IoT technology, it provides a unique solution to the difficulties of deploying an EVCS in a home environment. The EVCS is easy for EV owners to plug in and use, and it can be put in either an indoor or outdoor setting. A multiple-output charging device, a Raspberry Pi controller with Wi-Fi, and a simple mobile app are the core elements of this setup. The charging unit is connected to the Raspberry Pi controller, which then relays charging status and power usage information to the user's mobile device or an online server.

Keywords—Electric vehicle, Internet of things, Charging station, Raspberry Pi and Motor control

I. INTRODUCTION

Considering the rising demand for EVs, the automobile sector is witnessing a dramatic shift. The shift from automobiles powered by internal combustion engines to those powered by electricity is urgently needed as worries about global warming and environmental sustainability increase. However, the widespread usage of EVs depends on the development of a convenient and inexpensive charging infrastructure. Existing EV charging solutions have their own set of restrictions and obstacles that might discourage prospective EV owners and slow the rate of EV adoption in the residential sector, such as high installation costs, limited access, and an absence of remote management capabilities.

Toxic chemicals are released into the atmosphere by the exhaust of a traditional internal combustion engine (ICE), posing a threat to human respiratory health. The primary barrier to deployment and acceptance of EVs is the length of time required to charge using conventional AC chargers. Electric vehicle manufacturers have responded to these problems by introducing DC charging stations. DC charger

issues are consistent with the battery pack's health and safety risks caused by fast charging. IoT-integrated facilities will need to undergo an efficiency and safety paradigm change in order to break through the previous obstacles.

Charging stations, as a kind of EV infrastructure, serve an increasingly crucial role in the expansion of the EV market. The impact of the point of charging on the electricity grid may be ignored in the early stages of EV adoption. The potential influence on the grid, particularly the load profile of the grid, would continually arise as EV penetration rises, as will the demand for charging and the amount of charging stations. Currently, predicting charging station demand is dependent on theoretical approaches such as the combination of historical load curve and valley and other techniques. Mathematical statistical methods, applied to real-world measurement data, will provide the most useful fitting curve results.

The load curve of the grid demonstrates demand from consumers, so rationally forecasting the load curve plays an important role in energy dispatching and operating. The impact of charging stations on the load curve of the power grid depends heavily on the charging time of batteries and the charging mode. This research aims to solve these major issues by suggesting the creation of a low-priced, easy-to-use EVCS that not only allows for fast, easy charging but also includes remote surveillance and payment processing. The need for a stable and focused user charging infrastructure is becoming more pressing as the demand for EVs rises. To overcome these problems, a system is created that uses the IoT and is controlled by a Raspberry Pi.

The resultant system provides an innovative and comprehensive solution to the difficulties of placing EVCS devices in the house. With this technology, EV owners can simply plug in their cars and start the charging process with a few simple clicks. Its solution adaptability and usability are further strengthened by the fact that it may be used in a wide range of contexts, both inside and out. Core components of this modern EVCS system include a multi-output charging unit, a Raspberry Pi controller with Wi-Fi, and a pleasant mobile application. A user's smartphone or a centralized internet server receives essential information regarding

charging progress and electrical consumption via the Raspberry Pi controller, which is connected to the charging unit.

II. LITERATURE SURVEY

The incorporation of IoT technology into the design of home EVCS is a major step forward in meeting the hurdles of EV uptake in the residential market. To better understand the challenges and potential solutions associated with household EV charging infrastructure, this section will examine the relevant literature and current projects. Having a reliable and easily accessible charging infrastructure, especially in residential areas, is crucial to the widespread adoption of EVs [1]. The high upfront costs of installing charging infrastructure and the scarcity of available charging ports have been cited as barriers to EV adoption by homeowners in previous research. These obstacles are known to be slowing factors in the home EV adoption rate.

As the population rises exponentially, as well as the number of cars on the road. It is essential to switch to a new kind of transport, and EVs are a fantastic illustration of this since it is well-known that the planet has a limited amount of petroleum. Most EVs are charged by plug-in charging, which requires the user to physically connect a charging cable to the vehicle. When utilizing wireless charging, users don't have to do anything to the plug. Thus, less human interaction is required, and the risk of electric shock from linked linkages is reduced. Plug-in EVs have a short range and must use heavy, cumbersome batteries [2]. The primary benefits of wireless charging technologies include improved driving range and decreased battery size. It cuts down on both the time and money needed to charge the vehicle. As a result, the environmental and economic advantages of EVs grow, and their widespread adoption accelerates. Since charging takes so much time, it's more cost-effective to do it when the vehicle is parked. Therefore, it is more economical to integrate user-friendly charging and parking facilities that use IoT technology [3]. The cloud and mobile devices may both be used to store and access data.

With real short-term economic advantages and the benefits of being less vulnerable to oil price volatility, electric cars may also aid in lowering the oil bill. Assuming a decent EV range of 25–30 km and an average charging time of one hour, this would make electric vehicles a viable option for 78 percent of drivers who travel less than fifty kilometers daily [4]. The battery size and powertrain complexity of electric cars are the primary factors in determining their range. Recent studies' focus has been on finding ways to increase electric vehicle range. Therefore, products for controlling batteries and monitoring are essential. In [5], a comprehensive examination of the ways in which Li-ion batteries power EVs. Regarding high-capacity batteries that need a fast charging method, several different types of Li batteries are described and suggested, along with their basic working concepts, assembly, and performance.

Knowing the battery's state of charge (SOC) is crucial for charging and discharging it safely, which in turn increases the battery's lifespan. It may represent the ratio that exists between the battery's rated capacity and its balancing relationship in this manner. Therefore, SOC helps with power management. One of the most important measures of a battery is its SOC, which stands for its usable capacity. Battery load features were only some of the many variables that may affect SOC

[6]. The SoC measures how much of a battery's total capacity may still be charged. Like the gas gauge in a gas-powered automobile, the SoC of a battery may be used to determine how much energy is still in the battery to power an electric vehicle. Accurately calculating battery SoC aids in providing real-time data on the battery's exceptional capacity and energy, and it also guarantees a trustworthy and secure vehicle procedure. However, batteries are complex electrochemical devices whose SoC is difficult to determine since their performance is nonlinear and varies in response to both internal and external factors [7].

There are several techniques for calculating a battery's SOC. These techniques may be broken down into three categories: An electrochemical, adaptive, or electrical system is the third kind. Extremely precise electrochemical methods, however, need the use of electricity. The chemical composition of the battery makes it challenging to deploy software. Adaptive battery procedures need similar circuit architecture and a solution strategy [8]. The success of an adaptive method depends on how well the accompanying model works. In contrast, electrical methods need observable parameters like charging and discharging current, voltage at the terminal, and internal resistance. Among the most used methods for estimating SOC, the coulomb measuring methodology is popular since it is straightforward and easy to apply [9].

IoT has been an innovator in the EV charging infrastructure industry in recent years. Charging stations can be monitored and managed remotely in real time because of their internet connectivity. Although the IoT has been studied extensively for its potential to improve the performance and convenience of EV charging stations, its usage in private residences has received fewer resources [10]. It is impossible to overestimate the significance of user-friendliness in EVCS. Research indicates that user-friendliness plays a crucial role in whether or not a household would adopt an electric car. Attracting more home buyers to EVs requires a number of improvements, including more intuitive user interfaces, easier plug-and-charge procedures, and mobile apps for monitoring and management [11].

Different homeowners may have different ideas about where their electric vehicle charging stations should be installed. There is a requirement for charging solutions that can be deployed inside and outside in response to a wide variety of dwelling kinds and layouts, according to the available research [12]. The Raspberry Pi, a cheap and flexible single-board computer, has been used in a number of Internet of Things implementations. Data processing, device management, and internet connectivity make it an attractive option for managing and regulating electric vehicle charging infrastructure [13]. The feasibility of using Raspberry Pi controllers in this setting has been shown by previous research.

Streamlining the invoicing and payment process is a further area of importance for home EV charging. According to studies, more individuals would be willing to use EVs if payment mechanisms were more streamlined and convenient for users [14]. However, it is still difficult to implement these capabilities into home charging networks. The move toward electric cars is intrinsically linked to green values. Electric vehicles may help lessen the need for fossil fuels and the pollution they produce. To achieve these environmental objectives, it is essential that a sustainable household charging

infrastructure be developed [15]. In order to make the use of EVs more feasible for homes, this research expands on these results by offering a holistic approach that incorporates these features.

III. PROPOSED SYSTEM

The suggested solution intends to remove the obstacles that now stand in the way of the widespread adoption of home charging stations for EVs. The IoT, intuitive interfaces, flexible deployment choices, and an efficient payment processing system all combine together in this novel approach to provide a charging station that is both affordable and easily accessible. The Raspberry Pi controller is the brains of the operation, coordinating all of the parts and ensuring their proper functioning. Figure 1 presents the general layout of the PV-powered residential system for EV charging stations.

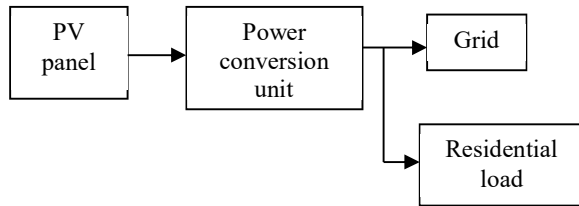


Fig. 1. The general layout of PV-powered residential application

This unique household charger system plugs into the wall and uses solar energy to power the house, charges the vehicle in both directions and connects to the internet of things. IoT and a Raspberry Pi 3 power a charging point, smart meter, and smart grid. The MPPT technology is used for solar power generation. The EV charging unit is the main part of the charging system that supplies electricity to the plugged-in EV. The unit's several outputs for various charging connections make it suitable for use with a wide variety of EV makes and models. The EVCS is controlled by a Raspberry Pi single-board computer. It handles data transfer between parts, data collection and processing, and billing regulation. The Raspberry Pi can connect to the internet and communicate with other devices because of its built-in Wi-Fi functionality. To make the charging procedure more manageable for EV users, a user-friendly smartphone app is being created. Users can safely manage purchases, begin charging, examine power use, and more, all from inside the app.

The IoT unifies disparate systems and enables two-way communication in near-real time. The EVCS can be effectively managed thanks to remote monitoring, control, and data collecting. Data interchange with the mobile app and IoT connection are both handled by the Raspberry Pi controller and are connected to the EV charging device. The system is a reliable and easily accessible option for household EV charging since it is capable of collecting and analyzing real-time data, offer an attractive interface, and process payments. This holistic strategy is meant to promote sustainability and efficiency for homeowners while also encouraging wider use of EVs.

There are three distinct types of charging methods available today: normal charge, fast charging, and battery replacement. The most common charging technique is the typical, slow way of plugging into an outlet in a personal space. It's the go-to method for families, and it may take anywhere from 5 to 8 hours to fully charge, depending on the strength of the battery. While Standard Charging is the most

efficient and safest charging method, it often takes too long to complete a charge. Emergency charging, which happens quickly, is essential. Quick charging, with a charging current of 150 to 400A, allows for rapid servicing in as little as 20 minutes to two hours.

The best approach to charge a device is by replacing the battery. Central charging at night will save a lot of money and time for those who utilize a reasonably inexpensive charging option during the day, but their batteries drain completely throughout the day. In addition, it lessens the toll on electricity and helps equalize the Peak-Valley Difference. Batteries in transportation vehicles must be regularly serviced. Fast and efficient battery removal is possible at charging stations. The Raspberry Pi is a small, nevertheless fast processor used as a central controller. The Raspberry Pi is a line of inexpensive single-board computers developed by the Raspberry Pi Foundation in order to improve the education of software engineering fundamentals at K-12 institutions and in nations with limited resources. The initial model's success exceeded expectations, and it is now being used in fields outside of the initial purpose of the product.

The Raspberry Pi does not come with any extra accessories like a keyboard, mouse, or case. However, certain creative elements have been added to several formal and casual packs. The suggested system uses a technology known as the IoT to track and share data on the health of charging stations. Time-of-use Pricing (TOU) is another name for the time-based tariff under which the controller operates. The capacity of EV batteries, the design of charging facilities, and driving behavior were analyzed for their effects on load profiles. The length of time between charges is dependent on the size of the EV's battery. Simply said, the higher the capacity, the longer it will take between charges. The ability to charge is connected to charging power as well. The overall distribution of time for charging will also be impacted by the availability of charging stations.

Most photovoltaic systems have components like solar modules to generate electricity, a battery charger to convert board output to battery voltage, a battery bank to store energy during the day and release it at night, an inverter to convert battery voltage to the optimal line voltage for running home appliances, and a line source chooser that switches between using solar power and the power grid. The inverter, which takes 48V DC and converts it to normal line voltage, is powered by the batteries. Inverters with a 5KW continuous rating can power almost all household appliances, including a clothes dryer, washing machine, dishwasher, and electric stove. The battery current might reach 200A while the inverter is supplying a large load. IoT technology is used for data collection, monitoring, and dissemination in the suggested system.

The real-time data regarding the system management and performance can be extracted by a sensor multiplexer with network topology inclusion of IoT-enabled devices in EV chargers. It is important to recognize the security problems that must be addressed when overseeing the integration and implementation of IoT-enabled DC charging infrastructure. To provide reliable, space-saving, and independent DC power sources for EVs with intelligent fast-charging capabilities, the IoT protocols must be flexible. The IoT is a network that connects devices, automobiles, appliances, and other physical objects equipped with software, sensors, actuators, electronic components, and a computerized system, allowing them to

communicate with one another and share data. Everything is easily recognized thanks to its built-in registration system, but it also functions inside the existing Internet infrastructure.

IV. RESULTS AND DISCUSSION

Integration of EVs into residential settings has gained considerable traction in the efforts to advance sustainable mobility options. In order to make this change, knowing

characteristics like current and voltage status is essential for the creation of effective and user-friendly domestic EVCSs. Insight into the electrical properties of the charging infrastructure for electric vehicles is presented here via the results of tests conducted at an individual charging station. Figure 2 shows the PV-powered grid station for residential applications in MATLAB.

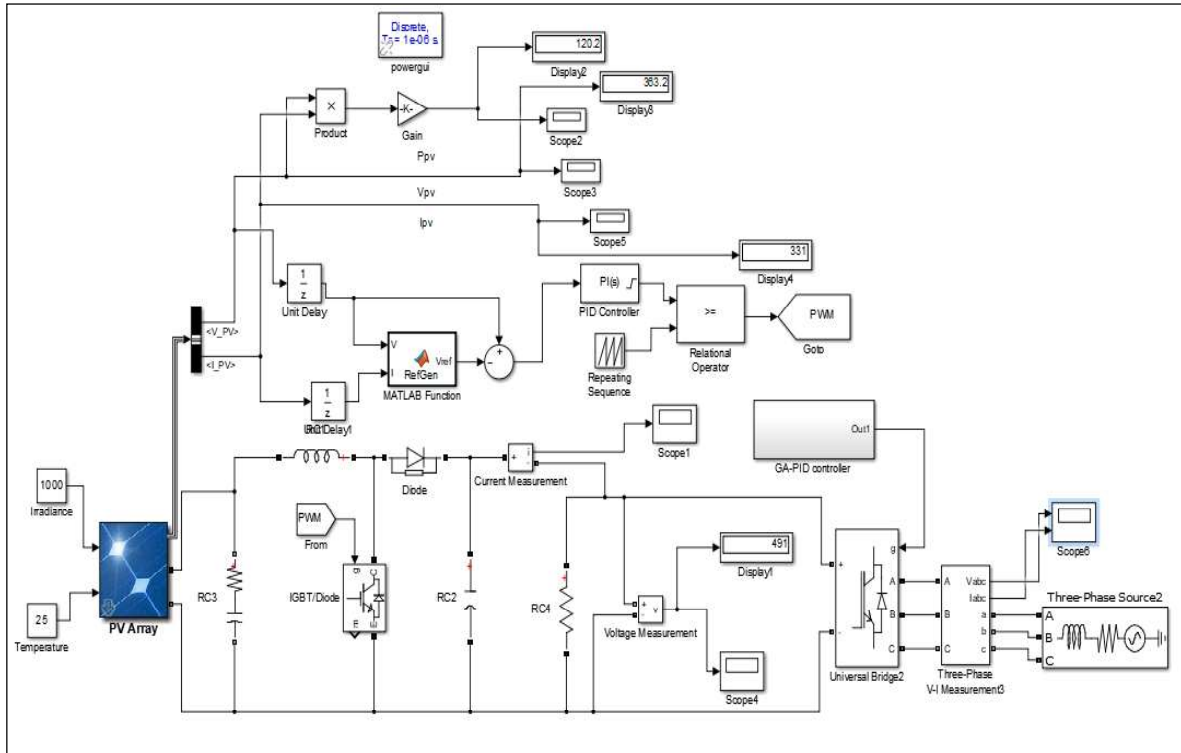


Fig. 2. Matlab simulation of PV grid system

Charging is a dynamic process, as seen by the graph in Figure 3. The charging power first increases quickly, which might be an indication of a greater current being given to the EV at the start of charging. Therefore, power consumption varies within a range, which may be attributable to variables such as battery capacity, charging mode, or power supply changes. Changes in voltage and current might account for the observed oscillations in charging power. To have a deeper understanding of the charging process, it may be helpful to plot these values over time.



Fig. 3. Charging status of the system

The current and voltage status of the EVCS is shown in Figure 4. Over 18.5 Amperes (A) of current had been measured coming from the charging station. This measurement indicates the amount of electricity transmitted from the charger to the connected EV. The charging station has a voltage reading of 240 Volts (V). When it comes to charging, voltage is a crucial factor that affects both power production and efficiency. Charging is quick and easy because of the current's moderate speed and the voltage's efficiency. In a domestic setting, where convenience dominates, this meets the demands of EV owners who desire fast charging sessions.

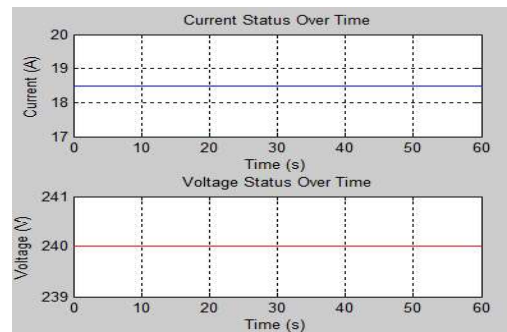


Fig. 4. Current and voltage status of EVCS

It's important to note that the data shown here depends on constant amounts of current and voltage. Fluctuations and variances are possible in real-world circumstances and must be well controlled to provide optimal charging performance. More research and analysis are needed to determine how these metrics affect user satisfaction, efficiency, and the environment as a whole.

V. CONCLUSION

A significant advance step in resolving issues connected with EV adoption in the home has been taken with the creation of an EVCS that makes use of IoT technology. The measurements of voltage and current status given in this research provide insight into the real-world relevance of such a system. According to the findings, the charging station provides a useful combination of current status (about 18.5 A) and voltage status (around 240 V). These numbers correspond to a reasonably quick charging rate and efficient power output, both of which aid in providing a fast, convenient, and easy EV charging experience. This is significant because it meets the expectations of EV owners who want to charge their vehicles quickly and easily without having to choose between various models and brands of EVs. The findings also show the potential of the IoT-enabled technology to give real-time data on current and voltage status as well as other elements of EV charging. The remote monitoring user experience and data analysis might all be improved with this data-driven strategy for running the charging station more efficiently.

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